USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[22TCSE239]

FOURTH SEMESTER B.Tech EXAMINATION, APRIL-2024

DATA EXPLORATION AND PREPARATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02 = 10**		
1.	**a.**	What is Data Science?	L-2	CO1	**02**
	b.	Explain data collection in the context of data science	L-2	CO2	**02**
	c.	Illustrate Exploratory Data Analysis.	L-3	CO3	**02**
	d.	What is a histogram? Explain its use in data science	L-2	CO4	**02**
	e.	Explain feature engineering in data exploration and preparation	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Illustrate different Software Tools used for Data Acquisition	L-3	CO1	**05**
		OR			
	b.	What are the key challenges encountered during the process of Data Exploration and Preparation?	L-2	CO1	**05**
3.	**a.**	What are the distinctions between Normalization and Standardization techniques in data preprocessing?	L-2	CO2	**05**
		OR			
	b.	Describe the core steps for basic data cleansing using Python programming	L-2	CO2	**05**
4.	**a.**	Explain Best practices for achieving Effective Visualization	L-2	CO3	**05**
		OR			
	b.	Illustrate the use of plotly library in data exploration	L-3	CO3	**05**
5.	**a.**	Explain analysis of dataset using Measures of Central Tendency methods	L-4	CO6	**05**
		OR			
	b.	Analyze the working of Spearman Rank Correlation Coefficient with the help of dataset X=[10,5,8,3,7] y=[20, 15,18,12,17]	L-4	CO6	**05**
6.	**a.**	What are Filter Methods? Explain the Characteristics of Filter Methods	L-2	CO5	**05**
		OR			
	b.	Explain Image Data Feature Extraction with an example.	L-3	CO5	**05**

7.	**a.**	Explain the types of Data Sources with an example.	L-3	CO1	**05**
		OR			
	b.	Explain various methods to handle missing data in data cleaning process	L-3	CO2	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Explain different Data Collection methods in data preparation	L-2	CO1	**10**
		OR			
	b.	Elaborate the data science process with neat diagram.	L-3	CO1	**10**
9.	**a.**	List and Explain the different methods to identify the outliers in the dataset	L-3	CO2	**10**
		OR			
	b.	Explain different types of data transformation techniques	L-3	CO2	**10**
10.	**a.**	Elaborate different types of data visualization techniques	L-4	CO3	**10**
		OR			
	b.	How to choose the right graph for data, what are the points to be considered for the same?	L-2	CO3	**10**
11.	**a.**	Illustrate Linear Regression with suitable example	L-3	CO4	**10**
		OR			
	b.	Demonstrate the working of K-Means and PCA Methods	L-3	CO4	**10**
12.	**a.**	Demonstrate the working of DBSCAN and Mean Shift Clustering Methods	L-3	CO4	**10**
		OR			
	b.	Explain Wrapper Methods in feature selection.	L-2	CO5	**10**
13.	**a.**	Explain embedded methods in feature selection	L-2	CO5	**10**
		OR			
	b.	Illustrate Text Data Feature Extraction with suitable dataset	L-3	CO5	**10**